

Ex No: 10	INSTALL AND SET UP A NOSQL DATABASE (E.G., MONGODB). WRITE QUERIES TO INSERT, UPDATE, AND QUERY DATA IN MONGODB
Date :	

AIM:

To install and set up a NoSQL database (e.g., MongoDB). Write queries to insert, update, and query data in MongoDB

Step 1: Install MongoDB**Download MongoDB:**

Visit the MongoDB Download Center.

Select the appropriate version for your operating system and download the installer.

Install MongoDB:

Follow the installation instructions specific to your operating system.

Start MongoDB:

Open a terminal or command prompt.

Start the MongoDB server:

On Windows: C:\Program Files\MongoDB\Server<version>\bin\mongod.exe

On macOS/Linux: Run mongod command.

Open MongoDB Shell:

Open a new terminal or command prompt.

Start the MongoDB shell by running mongo.

Step 2: Create a Database and Collections

```
> use library
```

switched to db library

Step 3: Insert Data

```
> db.authors.insertMany([
```

```
{ "AuthorID": 1, "FirstName": "George", "LastName": "Orwell" },
```

```
{ "AuthorID": 2, "FirstName": "Aldous", "LastName": "Huxley" },
```

```
{ "AuthorID": 3, "FirstName": "J.K.", "LastName": "Rowling" } ]])
```

Output

```
{ "acknowledged": true, "insertedIds": [ 1, 2, 3 ] }
```

```
> db.books.insertMany([
```

```
{ "BookID": 1, "Title": "1984", "Genre": "Dystopian", "PublicationYear": 1949, "Authors": [1] },
```

```
{ "BookID": 2, "Title": "Brave New World", "Genre": "Dystopian", "PublicationYear": 1932, "Authors": [2] },
```

```
{ "BookID": 3, "Title": "Harry Potter and the Sorcerer's Stone", "Genre": "Fantasy", "PublicationYear": 1997, "Authors": [3] } ]])
```

Output

```
{ "acknowledged": true, "insertedIds": [ 1, 2, 3 ] }
```

```
> db.borrowers.insertMany([
```

```
{ "BorrowerID": 1, "FirstName": "John", "LastName": "Doe", "MembershipDate": new Date("2023-01-01") },
```

```
{ "BorrowerID": 2, "FirstName": "Jane", "LastName": "Smith", "MembershipDate": new Date("2023-02-15") } ]])
```

Output

```
{ "acknowledged": true, "insertedIds": [ 1, 2 ] }
```

```
> db.borrowedBooks.insertMany([
```

```
{ "BorrowerID": 1, "BookID": 1, "BorrowedDate": new Date("2023-03-01"), "ReturnDate": new Date("2023-03-15") },
```

```
{ "BorrowerID": 2, "BookID": 3, "BorrowedDate": new Date("2023-03-05"), "ReturnDate": new Date("2023-03-20") } ]])
```

Output

```
{ "acknowledged": true, "insertedIds": [ 1, 2 ] }
```

Step 4: Update Data

```
> db.authors.updateOne( { "AuthorID": 1 }, { $set: { "LastName": "Smith" } } )
```

Output

```
{ "acknowledged": true, "matchedCount": 1, "modifiedCount": 1 }
```

Step 5: Query Data

```
> db.authors.find().pretty()
```

Output

```
{ "_id": ObjectId("..."), "AuthorID": 1, "FirstName": "George", "LastName": "Smith" },  
{ "_id": ObjectId("..."), "AuthorID": 2, "FirstName": "Aldous", "LastName": "Huxley" },  
{ "_id": ObjectId("..."), "AuthorID": 3, "FirstName": "J.K.", "LastName": "Rowling" }
```

Step 6: Delete Data

```
> db.authors.deleteOne({ "AuthorID": 1 })
```

Output

```
{ "acknowledged": true, "deletedCount": 1 }
```

RESULT

Thus the MongoDB was installed and database was created and various operations like insert, update, query, and delete data was executed in MongoDB